

# Site Risk Scoring Methodology

Network Planning, North-Central Region

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Version 1.2, 2026-08-15. Supersedes the ad-hoc risk-flag column previously maintained on the site tracker.

## 1. Purpose

Site acquisition for a rural rollout sits across four independent risk axes (paperwork, community engagement, milestone timing, vendor dependency) and a single qualitative Risk\_Flag column is no longer sufficient to triage the 45-site Phase 1 list. This methodology defines a composite risk score per site, computed from four numeric inputs and one weighted formula. The output is a numeric score plus a four-band qualitative label (LOW / MED / HIGH / CRITICAL) used in the weekly review.

## 2. Inputs (4 numeric columns on records\_nasarawa\_sites)

| Column                   | Range    | Definition                                                                                                                                                                                                                  |
|--------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Paperwork_Completion_Pct | 0 to 100 | Share of paperwork checklist items complete for this site (land title, environmental approval, traditional leader consent, LGA permit, community hearing minutes).                                                          |
| Leader_Engagement_Days   | integer  | Positive integer = days since last engagement. Zero = no engagement scheduled and none in the past 30 days. Positive small (1 to 7) = recent past engagement OR near-term scheduled engagement (use days_until).            |
| Milestone_Buffer_Days    | integer  | Days from focal moment to that site's intended commissioning date, measured relative to the Phase 1 milestone (currently Oct 12, 2026 post-revision). Negative = site commissioning has already slipped past the milestone. |
| Vendor_Dependency_Score  | 1 to 5   | Operational dependency on customs-cleared vendor equipment. 1 = self-contained civil works. 5 = depends on equipment held in customs as of the focal moment.                                                                |

## 3. Composite risk formula

```
composite_risk = (100 - paperwork_pct) * 0.35
+ max(0, 7 - leader_engagement_days_until) * 5 * 0.25
+ max(0, 14 - milestone_buffer_days) * 3 * 0.20
+ vendor_dependency_score * 4 * 0.20
```

Weights 0.35 / 0.25 / 0.20 / 0.20 reflect the empirical priority order: paperwork incompleteness dominates because it is the precondition for everything else; community engagement is second because a missed traditional-leader meeting can stop a site overnight; milestone buffer comes third because schedule slips are recoverable in most cases; vendor dependency is fourth because the operations team can compensate for short equipment delays.

## 4. Risk bands

| Band     | Score range | Operational meaning                                                              |
|----------|-------------|----------------------------------------------------------------------------------|
| LOW      | 0 to 20     | On track. Standard weekly review touch only.                                     |
| MED      | 21 to 50    | Needs attention this week. Owner flagged in Friday status.                       |
| HIGH     | 51 to 100   | Site is at risk of slipping the milestone. Mitigation plan required by next Wed. |
| CRITICAL | 101 plus    | Site cannot meet the milestone on current trajectory. Escalation to director.    |

## 5. Worked example: NSW-PHASE1-S03 (Akun ward, Lafia LGA)

S03 sits in the active community-objection bucket. The traditional leader, Chief Musa Adamu, signed a typed objection letter on Sept 28 proposing a town hall meeting for Oct 10. A subsequent phone call on Oct 2 between Chief Musa and Ifeanyi Okeke (FC) rescheduled the meeting to Oct 7, which is 3 days from the focal moment. Field paperwork sits at 15 percent complete because the environmental approval and the LGA permit are both pending review.

Inputs:

```
Paperwork_Completion_Pct = 15
Leader_Engagement_Days = 3 (Oct 7 meeting; days_until = 3)
Milestone_Buffer_Days = -5 (site commissioning will slip Oct 17, past the Oct 12 milestone)
Vendor_Dependency_Score = 4 (Akun cluster depends on the 2 customs-held radio units)
```

Composite:

```
(100 - 15) * 0.35 = 29.75
max(0, 7 - 3) * 5 * 0.25 = 5.00
max(0, 14 - (-5)) * 3 * 0.20 = 11.40
4 * 4 * 0.20 = 3.20
Total = 49.35 -> rounds to 49.4 -> MED band (right at the edge of HIGH).
```

Operational read: S03 is one missed leader meeting away from HIGH. If Oct 7 slips, leader\_engagement\_days\_until becomes 0 and the penalty jumps from 5.00 to 8.75, pushing composite into the high 50s. Keep the Oct 7 meeting at all costs.

## 6. Weekly review usage

Run the formula across all 45 Phase 1 sites every Sunday evening. Report sites in score-descending order, calling out:

- (a) every HIGH and CRITICAL site by Site\_ID with current score and the single highest-weight input,
- (b) every MED site within 5 points of HIGH,
- (c) every site whose score has increased by more than 10 points week-over-week (rate-of-change check).

The full ranked table goes into the source log for the weekly review.

## 7. Limitations

The formula assumes the four inputs are independent, which is not strictly true: low paperwork and low leader engagement often co-vary because both reflect a stalled community engagement track. The formula tolerates this co-variation by capping the leader penalty at 8.75 (i.e. saturates at days\_until = 0), preventing double-counting. The vendor dependency input is binary in practice (a site either needs the customs-held radios or it does not), but kept as 1-5 to allow for partial-dependency scoring as the operations team standardises on a per-line equipment-allocation matrix.

- end of methodology -

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